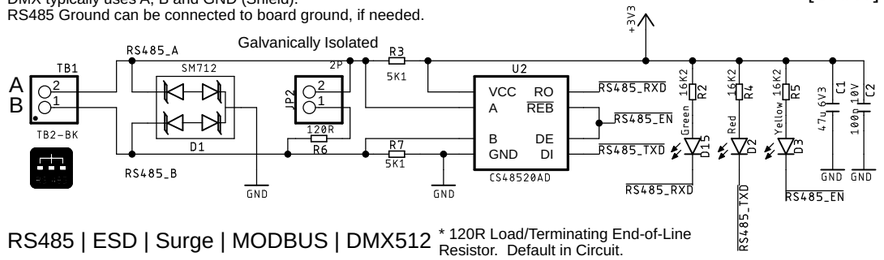
RS485 typically only uses A and B. Unless shielded cables are used.
DMX typically uses A, B and GND (Shield).
RS485 Ground can be connected to board ground, if needed.

[UART]



RS485 | ESD | Surge | MODBUS | DMX512

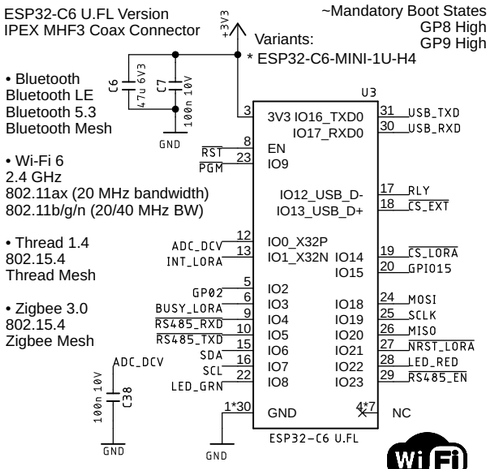
* 120R Load/Terminating End-of-Line Resistor. Default in Circuit.

ESP32-C6 U.FL Version
IPEX MHF3 Coax Connector

~Mandatory Boot States
GP8 High
GP9 High

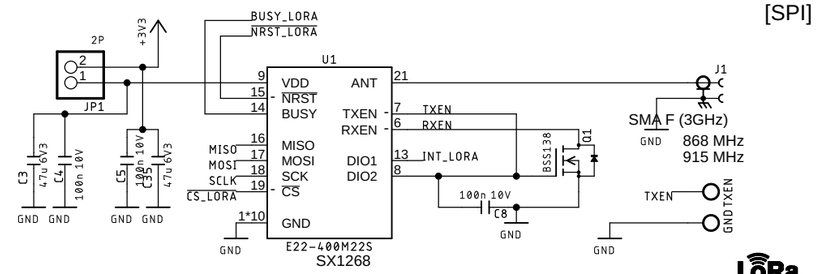
Variants:
* ESP32-C6-MINI-1U-H4

- Bluetooth
  Bluetooth LE
  Bluetooth 5.3
  Bluetooth Mesh
- Wi-Fi 6
  2.4 GHz
  802.11ax (20 MHz bandwidth)
  802.11b/g/n (20/40 MHz BW)
- Thread 1.4
  802.15.4
  Thread Mesh
- Zigbee 3.0
  802.15.4
  Zigbee Mesh



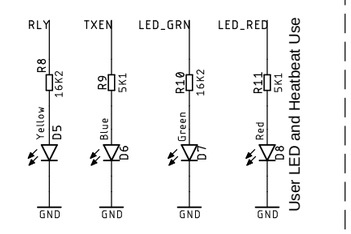
MCU | ESP32-C6

[SPI]



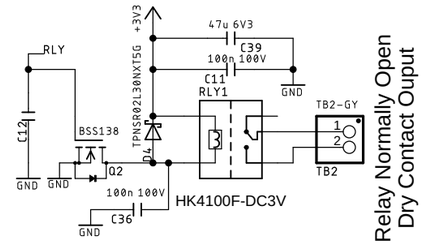
LoRa | EByte E22-400M22S | SX1268

410 MHz to 493 MHz ISM Band



Status | LED

User LED and Heatbeat Use

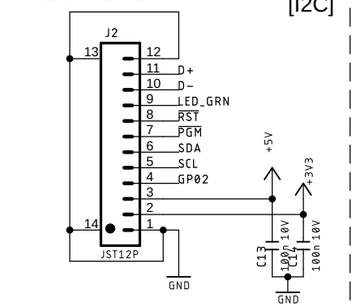


Relay Output (Galvanically Isolated)

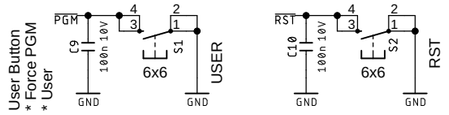
Relay Normally Open
Dry Contact Ouput

JST-12 SH VERT 1.0MM

[I2C]



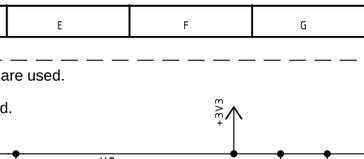
Expansion Interface | OLED (3V3)



Tactile Switches | Reset | Usr/Pgm

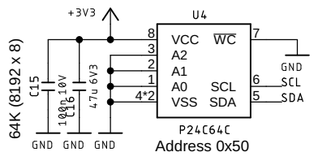
Appx DCV/ADC

05 V - 0.06 V
12 V - 0.16 V
20 V - 0.27 V
30 V - 0.40 V
40 V - 0.54 V
50 V - 0.68 V
60 V - 0.81 V
70 V - 0.95 V
75 V - 1.01 V

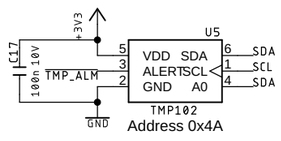


ADC DC Volts

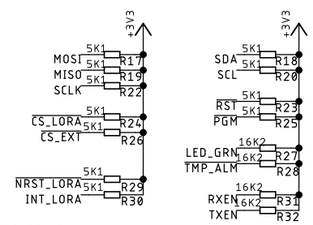
[I2C]



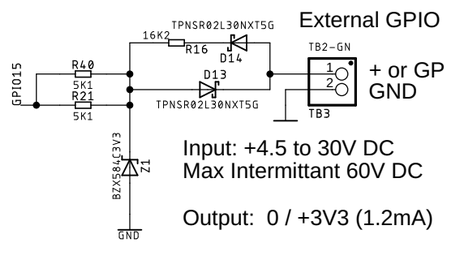
EEPROM | 24C64



Temperature | TMP102



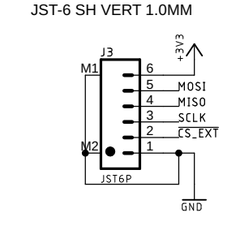
GPIO Pullups



GPIO15 | GPIO Input or Output

JST-6 SH VERT 1.0MM

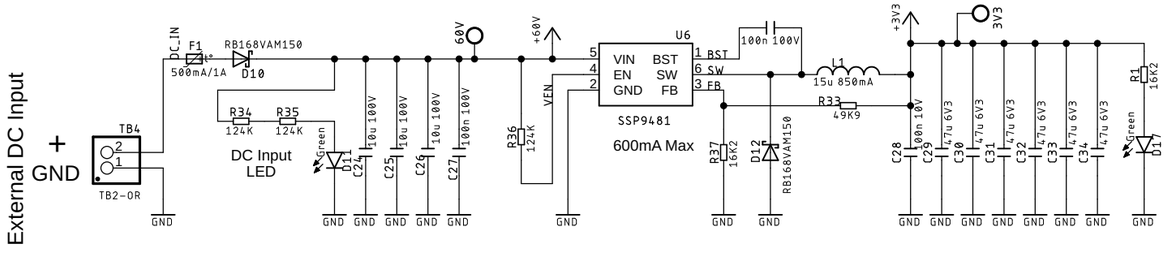
[SPI]



SPI Expansion | MicroSD (3V3)

! Beware of any High Voltages

DC In 5-60V. Maximum 75V Input.
Permanant damage WILL occur > 75V !
* Reverse Polarity Protected
* Fuse Hold 500mA Trip 1A Resettable

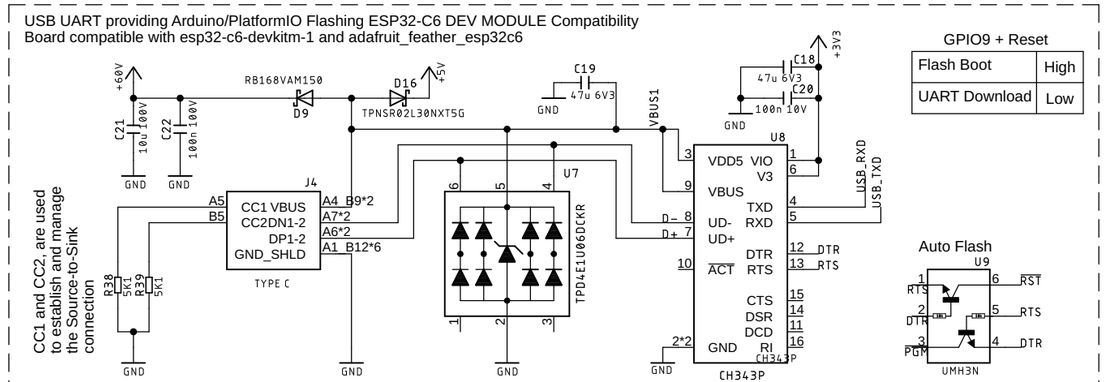


High Voltage Power Suppy | HVDC SMPS 3V3 DC-DC

USB UART providing Arduino/PlatformIO Flashing ESP32-C6 DEV MODULE Compatibility
Board compatible with esp32-c6-devkitm-1 and adafruit_feather_esp32c6

GPIO9 + Reset

Flash Boot	High
UART Download	Low



USB | USB UART | Serial Monitoring | USB Flashing | Debug | ESD | 5V Power

External GPIO

Enclosure VG-P41F 80x54x32 Main PCB 79x49mm

For further information and code examples, please see our code and GitHub WIKI pages. ditronix.net | GitHub.com/DitroniX | Hackster.io/DitroniX

Components optimised for BOM

DitroniX
Home Automation | Internet of Things | Energy Monitoring

Production

(C)2021-2026 - DitroniX - Dave Williams G8PUO

ESPKite LoRa Controller

STEM

S T E M

ESP32-C6 SX1268 SDK **ditronix.net** DitroniX Ltd – UK	ESPKite ESP32C6 E22-400M22S SX1268 1.2606.104	
	ESPKITE-LORA-ESP32-C6-RS485-CONTROLLER-SDK	
	Sheet: 1/1	Drwg/Chk: **Dave Williams** 14/06/2026 12:05